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Student Number



Barker
College

2019

TRIAL HIGHER SCHOOL
CERTIFICATE EXAMINATION

Mathematics

Staff Involved:

AM FRIDAY 9TH AUGUST

BHC*	ARP*
DZP	PJR
GPR	ARM
AYG	JGD
ESP	JZT
RAS	

140 copies

General

Instructions:

- Reading Time - 5 minutes
- Working time - 3 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided separately
- In Questions 11 - 16 show relevant mathematical reasoning and/or calculations

Total marks:
100

Section I - 10 marks (pages 2 - 4)

- Attempt Questions 1 - 10
- Allow about 15 minutes for this section

Section II - 90 marks (pages 5 - 10)

- Attempt Questions 11 - 16
- Allow about 2 hours and 45 minutes for this section

Section I – Multiple Choice

10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1 – 10.

- What is the value of $f'(2)$ if $f(x) = 3x^4 - x$?
 - 46
 - 94
 - 95
 - 13823
- What is the minimum value of the function $f(x) = 4 + 2\sin 2x$?
 - 0
 - 2
 - 4
 - 6
- In an arithmetic sequence $T_{10} = 0$ and $T_{12} = 8$. What is the value of T_1 ?
 - 48
 - 44
 - 40
 - 36
- If $\log_a 5 = 1.2$ then, correct to one decimal place, what is the value of a ?
 - 3.7
 - 3.8
 - 3.9
 - 4.0

5. If $\cos \theta > 0$ and $\operatorname{cosec} \theta = -3$ what is the exact value of $\cot \theta$?

- A. $\frac{1}{\sqrt{8}}$
- B. $\frac{-1}{\sqrt{8}}$
- C. $\sqrt{8}$
- D. $-\sqrt{8}$

6. What is the value of $\sum_{r=1}^3 r3^{-r}$?

- A. $\frac{2}{3}$
- B. $\frac{19}{27}$
- C. $\frac{20}{27}$
- D. $\frac{7}{9}$

7. What is the equation of the directrix of the parabola $(y-2)^2 = -12(x+4)$?

- A. $x = -7$
- B. $y = -7$
- C. $x = -1$
- D. $y = -1$

8. Over the interval $a \leq x \leq b$ a function $y = f(x)$ is described as “increasing at a decreasing rate”. This means that, for all values of x such that $a \leq x \leq b$,

- A. $f'(x) > 0$ and $f''(x) < 0$
- B. $f'(x) < 0$ and $f''(x) > 0$
- C. $f'(x) > 0$ and $f''(x) > 0$
- D. $f'(x) < 0$ and $f''(x) < 0$

9. If $-\frac{\pi}{4} < \theta < \frac{\pi}{4}$, find the value of the limiting sum of $1 - \tan^2 \theta + \tan^4 \theta - \tan^6 \theta + \dots$

- A. $\sec^2 \theta$
- B. $\operatorname{cosec}^2 \theta$
- C. $\cos^2 \theta$
- D. $\sin^2 \theta$

10. What is $\int_{-\frac{\pi}{3}}^{\frac{\pi}{3}} -x^3 \sin x \, dx$ equal to?

- A. 0
- B. $2 \int_0^{\frac{\pi}{3}} -x^3 \sin x \, dx$
- C. $2 \left| \int_0^{\frac{\pi}{3}} -x^3 \sin x \, dx \right|$
- D. $\left| \int_{-\frac{\pi}{3}}^{\frac{\pi}{3}} -x^3 \sin x \, dx \right|$

End of Section I

Section II

90 marks

Attempt Questions 11 – 16

Allow about 2 hours and 45 minutes for this section.

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11 – 16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) [START A NEW BOOKLET]

(a) Solve $-x - 1 = \frac{4x + 11}{3}$. 2

(b) Express $\frac{6}{3 - 2\sqrt{5}}$ with a rational denominator. 2

(c) Find the values of x for which $|2x + 1| \leq 2 - x$. 2

(d) Differentiate $y = 2x \cos(2x)$. 2

(e) Evaluate $\lim_{x \rightarrow 2} \left(\frac{x^2 - 4}{2 - x} \right)$. 2

(f) Find $\int e^{4x+1} dx$. 1

(g) Find $\int \frac{x}{4 + x^2} dx$. 2

(h) Find $\int \tan x dx$. 2

End of Question 11

Question 12 (15 marks) [START A NEW BOOKLET]

(a) Evaluate $\frac{\ln(\pi^2)}{e^\pi}$ correct to 2 significant figures. 1

(b) Find the **exact** values of k for which the equation $2x^2 + kx + 25 = 0$ has two equal roots. 2

(c) Given that $\log_a b = 1.6$ and $\log_a c = 0.4$, find the value of $\log_a \left[(abc)^2 \right]$. 2

(d) The diagram shows triangle XYZ with sides $XY = 7\text{cm}$, $XZ = 10\text{cm}$ and $\angle XZY = 32^\circ$. 3

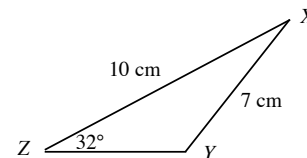


Diagram not drawn to scale

Note that $\angle XYZ > 90^\circ$

Calculate $\angle XYZ$ correct to the nearest degree.

(e)

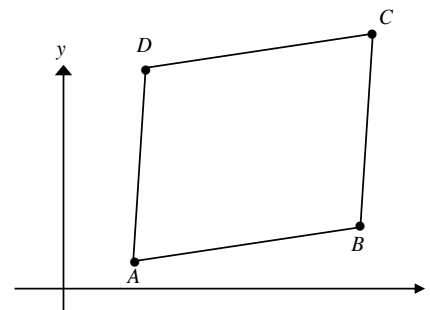


Diagram is not drawn to scale

In the diagram A , B , C and D are the points $(2, 1)$, $(7, 4)$, $(12, 11)$ and $(5, 6)$ respectively.

(i) Find the distance BD . 1

(ii) Find the midpoint of BD . 1

(iii) Show that AC is perpendicular to BD . 2

(iv) Find the midpoint of AC and hence give the most specific name of quadrilateral $ABCD$. 2

(v) Hence, or otherwise, find the area $ABCD$. 1

End of Question 12

Question 13 (15 marks) [START A NEW BOOKLET]

- (a) 108 tagged fish were released into a dam known to contain a lot of fish. 2
Later a sample of 62 fish was netted from this dam and then released.
Of these 62 fish in the sample it was recorded that 8 were tagged.

Estimate the total number of fish in the dam now.

- (b) (i) Prove that the line $y = x - 2$ is a tangent to the parabola $y = x^2 - 7x + 14$. 2

- (ii) Let T be the point where $y = x - 2$ touches the parabola $y = x^2 - 7x + 14$. 1
Show that the equation of the normal to the parabola at T is $y = -x + 6$.

- (iii) Find the area of the region enclosed between the parabola and the line 3
 $y = -x + 6$.

- (c) Mr Peattie brought to class a box of 30 chocolates which all appeared to be exactly alike. However, 22 chocolates were rambutan flavoured and 8 were durian flavoured. Lucy ate 3 chocolates chosen at random from the box. Find the probability that

- (i) the first chocolate Lucy eat was rambutan flavoured. 1

- (ii) Lucy ate 3 rambutan flavoured chocolates. 1

- (iii) Lucy ate at least one rambutan chocolate. 1

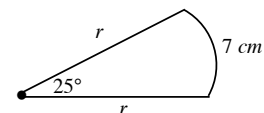
- (iv) Lucy ate exactly one rambutan chocolate. 1

- (d) Find the **exact** solutions of the equation $\ln x - \frac{3}{\ln x} = 2$. 3

End of Question 13

Question 14 (15 marks) [START A NEW BOOKLET]

- (a) Find the radius of this sector of a circle. 2
Give your answer correct to the nearest *cm*.



- (b) (i) Draw the graphs of $y = 5 \cos x$ and $y = 2 - x$ on the same set of axes for 2
 $-2\pi \leq x \leq 2\pi$.

- (ii) Using your graph in (i), how many solutions are there to the equation 1
 $5 \cos x = 2 - x$?

- (c) Sean is arranging rows of centicubes flat on a table. The centicubes are stacked in horizontal rows, with each row containing 2 less centicubes than the row below it.

There are 16 centicubes in the top row (1st row), 18 in the row below that (2nd row), and 20 in the row below that (3rd row), and so on.

There are n rows altogether and Sean always completes each row.

- (i) Write down an expression for the number of centicubes in the n th-row. 1

- (ii) Show that there are $n^2 + 15n$ centicubes altogether in the construction. 1

- (iii) If Sean says that there is a total of 448 centicubes in the construction, explain 1
why he must have miscounted.

- (iv) Sean has only 1100 centicubes from which to build the construction. What is 2
the maximum number of complete rows the construction can contain?

- (d) Sketch the curve $y = 4^{x-2}$ between $x = 2$ and $x = 4$.

- (i) Use the Trapezoidal Rule with 3 function values to estimate the area between 2
the curve, the x -axis and the lines $x = 2$ and $x = 4$.

- (ii) If this area is rotated about the x -axis, use Simpson's Rule with 3 function values 3
to approximate the volume of the solid that is formed.

End of Question 14

Question 15 (15 marks) [START A NEW BOOKLET]

- (a) A function is given by $f(x) = 3x^4 + 4x^3 - 12x^2$.
- (i) Find the coordinates of the stationary points and determine their nature. **3**
- (ii) Hence, sketch the graph of the function. **2**
- (iii) For what values of x is $f(x)$ increasing? **1**
- (iv) For what values of k will the equation $3x^4 + 4x^3 - 12x^2 + k = 0$ have no solution? **1**
- (b) The gradient of a curve is given by $\frac{dy}{dx} = 1 - 6\sin(3x)$. **2**
- The curve goes through the point $(0, 10)$.
- What is the equation of the curve?
- (c) (i) Assuming that $a > 1$, simplify $\int_0^{\ln a} e^{-2x} dx$, leaving your answer in the form of a simplified single fraction. **3**
- (ii) Find $\lim_{a \rightarrow \infty} \left[\int_0^{\ln a} e^{-2x} dx \right]$. **1**
- (d) Find $\int \frac{e^{-x}}{e^{-x}(1+e^x)} dx$. **2**

End of Question 15**Question 16** (15 marks) [START A NEW BOOKLET]

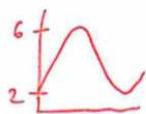
- (a) A “Barker Thirst Buster Water Bottle” is a closed cylinder of height h cm and radius R cm.
- (i) The capacity of the bottle is 1 litre. **1**
Find an expression for h in terms of R .
- (ii) Show that the surface area (S) of the bottle is given by $S = \frac{2}{R}(1000 + \pi R^3)$. **1**
- (iii) If the area of the metal used to make the bottle is minimised, find the height and radius of the bottle (to the nearest mm). **3**
- (b) Gen, Tia and Arjun each chose a number at random from 1 to 10 (inclusive).
Find the probability that:
- (i) they each chose the same number. **1**
- (ii) they each chose a different number. **1**
- (iii) two of them chose the same number and one of them chose a different number. **1**
- (iv) both the numbers chosen by Gen and Tia are greater than the number chosen by Arjun. **2**
- (c) Triangle ABC is right angled at B .
 D is a point on AC such that BD is perpendicular to AC . Let $\angle BAC$ be θ .
Draw a diagram showing this information. It is known that $\sqrt{3}BC + 6AD = 3AC$.
- (i) Show that $\sqrt{3} \tan \theta + 6 \cos \theta = 3 \sec \theta$. **2**
- (ii) Deduce that $2 \sin^2 \theta - \frac{1}{\sqrt{3}} \sin \theta - 1 = 0$. **2**
- (iii) Find θ . **1**

End of Question 16**End of Paper**

1. (C) $f(x) = 12x^3 - 1$

$$f'(2) = 95$$

2. (B)



3. (D) $0 = a + 9d \therefore T_1 = -36 + 0$

$$8 = a + 11d = -36$$

$$\therefore 2d = 9$$

$$d = 4.5$$

$$a = -36$$

4. (B) $a^{1.2} = 5 \quad a = \sqrt[1.2]{5}$

$$\frac{\log 5}{\log a} = 1.2 \quad = 3.12$$

$$\log a = \frac{1.2}{3.12}$$

$$\log a = \frac{1.2}{3.12}$$

5. (D) $\frac{S}{T} = \frac{A}{C} \checkmark$



$$\cot \theta = -\sqrt{9}$$

6. (A) $\frac{1}{3} + \frac{2}{9} + \frac{3}{27} = \frac{2}{3}$

7. (C)



8. (A)

9. (C) $a = 1$
 $r = \tan^2 \theta \therefore S_0 = \frac{1}{1 + \tan^2 \theta}$
 $= \frac{1}{\sec^2 \theta}$
 $= \cos^2 \theta$

10. (B) $-x^3 \sin x$ is even function

11. a) $-3x - 3 = 4x + 11$

$$7x = -14$$

$$x = -2$$

b) $\frac{6}{3-2\sqrt{5}} \times \frac{3+2\sqrt{5}}{3+2\sqrt{5}} = \frac{18+12\sqrt{5}}{9-20}$
 $= \frac{18+12\sqrt{5}}{-11}$
 $= -\frac{12\sqrt{5}-18}{11}$

c) $|2x+1| \leq 2-x$

$$2x+1 \leq 2-x \therefore$$

$$3x \leq 1$$

$$x \leq \frac{1}{3} \quad -2x-1 \leq 2-x$$

$$x \geq -3$$

$$\therefore -3 \leq x \leq \frac{1}{3}$$

1. i) $u = 2x \quad v = \cos 2x$

$$u' = 2 \quad v' = -2 \sin 2x$$

$$y' = 2 \cos 2x - 4x \sin 2x$$

e) $\lim_{x \rightarrow 2} \frac{(x+2)(x-2)}{2-x}$

$$= \lim_{x \rightarrow 2} -(x+2)$$

$$= -4$$

f) $\int e^{4x+1} dx = \frac{e^{4x+1}}{4} + C$

g) $\int \frac{x}{4+x^2} dx = \frac{1}{2} \int \frac{2x}{4+x^2} dx$
 $= \frac{1}{2} \ln(4+x^2) + C$

h) $\int \frac{\sin x}{\cos x} dx = - \int \frac{-\sin x}{\cos x} dx$
 $= - \ln |\cos x| + C$

2. a) $= 0.0989$

$$= 0.099 \quad (2 \text{ sig figs})$$

b) $2x^2 + 11x + 25 = 0$

$$\Delta = b^2 - 4ac = 0$$

$$k = \pm \sqrt{200}$$

$$= \pm 10\sqrt{2}$$

c) $\log_a(abc)^2 = 2(\log_a a + \log_a b + \log_a c)$
 $= 2(1 + 1.6 + 0.4)$
 $= 6$

d) $\frac{\sin \theta}{10} = \frac{\sin 32}{7}$

$$\sin \theta = \frac{10 \sin 32}{7}$$

$$\theta = 49.2$$

but θ obtuse

$$\therefore \theta = 180 - 49.2$$

$$= 131^\circ$$

e) i) $BD = \sqrt{4+4}$
 $= \sqrt{8} = 2\sqrt{2}$

ii) $m_{BD} = \frac{12}{2}, \frac{10}{2}$
 $= 6, 5$

iii) $m_{AC} = \frac{10}{10}$
 $= 1$

$$m_{BD} = \frac{2}{-2}$$

$$= -1$$

$$m_{AC} \times m_{BD} = -1 \therefore AC \perp BD$$

$$v) M_{AC} = (7, 6)$$

diagonals intersect at 90° but
not each other
 $\therefore$ a kite

$$v) \text{Area} = \frac{d_1 d_2}{2}$$

$$AC = \sqrt{10^2 + 10^2}$$

$$= \sqrt{200}$$

$$\therefore \text{Area} = \frac{\sqrt{1600}}{2}$$

$$= 20u^2$$

$$3.a) \frac{f}{62} = \frac{108}{2c}$$

$$x = 837$$

$$b) i) x^2 - 7x + 14 = x - 2$$

$$x^2 - 8x + 16 = 0$$

$$(x-4)^2 = 0$$

1 root $\therefore y = x - 2$ is a tangent

$$ii) x = 4, y = 2$$

$$\therefore T(4, 2)$$

$$y' = 2x - 7$$

at $x = 4$ $m = 1$ $\therefore$ normal is -1

$$\therefore y - 2 = -x + 4$$

$$\therefore \text{normal } y = -x + 6$$

iii) Find points of intersection

$$x^2 - 7x + 14 = -x + 6$$

$$x^2 - 6x + 8 = 0$$

$$(x-4)(x-2) = 0$$

$$x = 4, 2$$

$$y = 2, 4$$

$$\therefore \text{Area} = \int_2^4 (x+6 - x^2 + 7x - 14) dx$$

$$= \int_2^4 (-x^2 + 6x - 8) dx$$

$$= \left[-\frac{x^3}{3} + 3x^2 - 8x \right]_2^4$$

$$= -\frac{16}{3} + \frac{20}{3}$$

$$= \frac{4}{3} u^2$$

$$c) i) \frac{22}{30} = \frac{11}{15}$$

$$ii) P(3C) = \frac{22}{30} \times \frac{21}{29} \times \frac{20}{28}$$

$$= \frac{11}{29}$$

$$iii) P(\text{at least 1}) = 1 - P(\text{none})$$

$$= 1 - \left(\frac{8}{30} \times \frac{7}{29} \times \frac{6}{28} \right)$$

$$= \frac{143}{145}$$

$$iv) P(\text{exactly 3}) = \left(\frac{22}{30} \times \frac{8}{29} \times \frac{7}{28} \right) 3$$

$$= \frac{22}{435} \times 3$$

$$= \frac{22}{145}$$

$$1.) \ln x - \frac{3}{\ln x} = 2$$

$$(\ln x)^2 - 2\ln x - 3 = 0$$

$$\text{let } u = \ln x$$

$$u^2 - 2u - 3 = 0$$

$$(u-3)(u+1) = 0$$

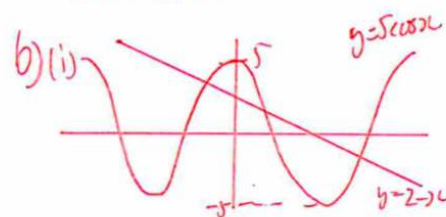
$$\therefore \ln x = 3 \text{ or } -1$$

$$x = e^3 \text{ or } e^{-1} \text{ or } \frac{1}{e}$$

$$14. a) 7 = r\theta$$

$$\theta = 0.436 \text{ rad}$$

$$\therefore r = 16.04$$



ii) 3 solutions

$$c) i) r_1 = 16 \quad r_2 = 18$$

$$a = 16 \quad d = 2$$

$$\therefore r_n = 16 + (n-1)2$$

$$= 14 + 2n$$

$$ii) S_n = \frac{n}{2} (2a + (n-1)d)$$

$$= \frac{n}{2} (32 + 2n - 2)$$

$$= \frac{n}{2} (30 + 2n)$$

$$= n^2 + 15n$$

$$iii) n^2 + 15n = 448$$

$$n^2 + 15n - 448 = 0$$

$$n = \frac{-15 \pm \sqrt{2017}}{2}$$

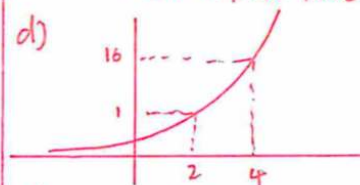
not a whole number $\therefore 448$
incorrect

$$iv) 15n + n^2 = 1100$$

$$n = \frac{-15 \pm \sqrt{4625}}{2}$$

$$= 26.5 \text{ as } n > 0$$

$\therefore 26$ complete rows



x	2	2.5	3	3.5	4
$f(x)$	1	2	4	8	16

$$A = \frac{0.5}{2} (1 + 16 + 2(2 + 4 + 8))$$

$$= 11.25 u^2$$

$$\begin{array}{ccc} x & 2 & 3 & 4 \\ f(x) & 1 & 4 & 16 \\ f(x)^2 & 1 & 16 & 256 \end{array}$$

$$V = \frac{\pi}{3}(1 + 256 + 4 \times 16)$$

$$= \frac{321\pi}{3} u^3$$

$$= 107\pi u^3$$

$$5. a) (i) f(x) = 12x^3 + 12x^2 - 24x = 0$$

$$x(x^2 + x - 2) = 0$$

$$x(x+2)(x-1) = 0$$

$$x = 0, -2, 1$$

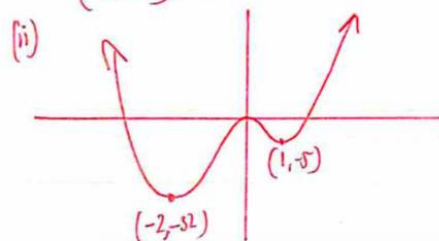
$$y = 0, -32, -5$$

x	-3	-2	-1	0	1/2	1	2
$f(x)$	-144	0	24	0	-7.5	0	96

$$\therefore (-2, -32) \text{ min}$$

$$(0, 0) \text{ max}$$

$$(1, -5) \text{ min}$$



$$(iii) \text{ Increasing } -2 < x < 0$$

$$x > 1$$

$$(iv) k > 32 \text{ for no solution}$$

$$b) y = x + 2 \cos 3x + C$$

$$\text{sub in } (0, 10)$$

$$10 = 0 + 2 + C$$

$$C = 8$$

$$\therefore y = x + 2 \cos 3x + 8$$

$$c) (i) \left. \frac{e^{-2x}}{-2} \right|_0^{\ln a}$$

$$= \frac{-1}{2a^2} + \frac{1}{2}$$

$$= \frac{a^2 - 1}{2a^2}$$

$$(ii) \lim_{a \rightarrow \infty} \left[\frac{a^2 - 1}{2a^2} \right]$$

$$= \frac{a^2}{2a^2}$$

$$= \frac{1}{2}$$

$$x) \int \frac{e^{-x}}{e^{-x}(1+e^x)} dx$$

$$= \int \frac{e^{-x}}{e^{-x} + 1} dx$$

$$= -\ln(e^{-x} + 1) + C$$

$$6 a) ii) V = \pi R^2 h$$

$$1000 = \pi R^2 h$$

$$h = \frac{1000}{\pi R^2}$$

$$(ii) SA = 2\pi R^2 + 2\pi R h$$

$$= 2\pi R^2 + \frac{2\pi R \times 1000}{\pi R^2}$$

$$= 2\pi R^2 + \frac{2000}{R}$$

$$= \frac{2}{R}(1000 + \pi R^3)$$

$$(iii) \frac{dS}{dR} = 4\pi R - \frac{2000}{R^2} = 0 \text{ for max/min}$$

$$\frac{4\pi R^3 - 2000}{R^2} = 0$$

$$4\pi R^3 = 2000$$

$$R^3 = \frac{500}{\pi}$$

$$R = \sqrt[3]{\frac{500}{\pi}}$$

$$= 5.42 \text{ cm}$$

$$= 54 \text{ mm}$$

$$R = 5 \quad \frac{dS}{dR} = -17$$

$$R = 6 \quad \frac{dS}{dR} = 19$$

$\therefore 54 \text{ mm}$ is a min

$$b) (i) P(\text{same}) = 1 \times \frac{1}{10} \times \frac{1}{10}$$

$$= \frac{1}{100}$$

$$(ii) P(\text{diff}) = 1 \times \frac{9}{10} \times \frac{8}{10}$$

$$= \frac{72}{100} = \frac{18}{25}$$

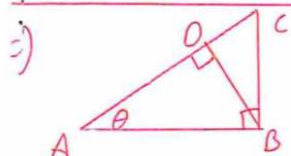
$$(iii) \left(1 \times \frac{1}{10} \times \frac{9}{10} \right) \times 3 = \frac{27}{100}$$

$$(iv) A \quad G \quad T$$

1	0.81	= 0.081
2	0.64	= 0.064
3	0.49	= 0.049
4	0.36	= 0.036
5	0.25	= 0.025
6	0.16	= 0.016
7	0.09	= 0.009
8	0.04	= 0.004
9	0.01	= 0.001

$$0.285$$

$$\text{or } \frac{57}{200}$$



$$\sqrt{3}BC + 6AD = 3AC$$

$$(i) \tan \theta = \frac{BC}{AB}$$

$$BC = AB \tan \theta$$

$$\cos \theta = \frac{AD}{AB}$$

$$AD = AB \cos \theta$$

$$\cos \theta = \frac{AB}{AC}$$

$$AC = AB \sec \theta$$

sub in BC, AD, AC

$$\therefore \sqrt{3} AB \tan \theta + 6 AB \cos \theta = 3 AB \sec \theta$$

$$\sqrt{3} \tan \theta + 6 \cos \theta = 3 \sec \theta$$

$$(ii) \frac{\sqrt{3} \sin \theta}{\cos \theta} + 6 \cos \theta = \frac{3}{\cos \theta}$$

$$\sqrt{3} \sin \theta + 6 \cos^2 \theta = 3$$

$$6(1 - \sin^2 \theta) + \sqrt{3} \sin \theta - 3 = 0$$

$$3 - 6 \sin^2 \theta + \sqrt{3} \sin \theta = 0$$

$$2 \sin^2 \theta - \frac{1}{\sqrt{3}} \sin \theta - 1 = 0$$

(iii)

$$\sin \theta = \frac{\frac{1}{\sqrt{3}} \pm \sqrt{\frac{1}{3} + 8}}{4}$$

$$= \frac{\frac{1}{\sqrt{3}} \pm \frac{5}{\sqrt{3}}}{4}$$

$$= \frac{3}{2\sqrt{3}} \text{ or } -\frac{1}{\sqrt{3}}$$

$$\theta = 60^\circ \text{ as } \theta > 0$$